

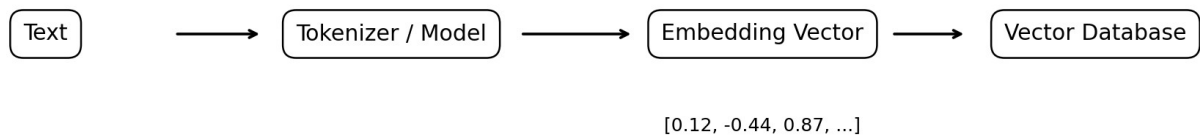
Embeddings

From text to meaningful numerical vectors

English definitions + Hinglish explanations + RAG examples

Learning goal

By the end of this guide, you will understand what an embedding is, why it is required, how semantic similarity works, and how embeddings are used inside a RAG pipeline.



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1. What is an Embedding?

Definition: An embedding is a numerical representation of data, such as text, images, audio, or products, in the form of a dense vector. The vector captures important semantic features and relationships in the original data.

Hinglish explanation

Embedding ka matlab hai kisi text ya object ko numbers ki list mein convert karna. Ye random numbers nahi hote. In numbers ke andar meaning, context aur similarity ki information encoded hoti hai.

Simple example

Input text	Illustrative embedding
Python is a programming language	[0.18, -0.42, 0.76, 0.11, ...]
Django is a Python web framework	[0.21, -0.39, 0.73, 0.15, ...]
Mango is a fruit	[-0.62, 0.31, -0.08, 0.84, ...]

Observation: The first two vectors are expected to be closer because both sentences discuss Python technology. The fruit sentence represents a different concept, so its vector is farther away.

2. Why Do We Need Embeddings?

Computers cannot directly understand meaning the way humans do. They process numbers. Embeddings create a bridge between human-readable information and mathematical operations.

- Semantic search: find information by meaning, not only exact keywords.
- Recommendation systems: identify similar users, products, movies, or songs.
- Clustering: group similar documents or customers automatically.
- Classification: use vector features to predict categories or labels.
- RAG: retrieve the most relevant document chunks for an LLM.
- Duplicate detection: identify semantically repeated or near-duplicate content.

Keyword search versus semantic search

Keyword search checks whether the same words are present. Semantic search checks whether the meaning is similar. For example, a search for "ways to reduce application latency" may retrieve a document about "improving API response time" even when the exact words differ.

3. How an Embedding Model Works

Input: The model receives a sentence, paragraph, image, or another supported data type.

Tokenization: Text is divided into smaller units called tokens.

Model processing: A trained neural network examines the tokens and their context.

Vector generation: The model outputs a fixed-length list of floating-point numbers.

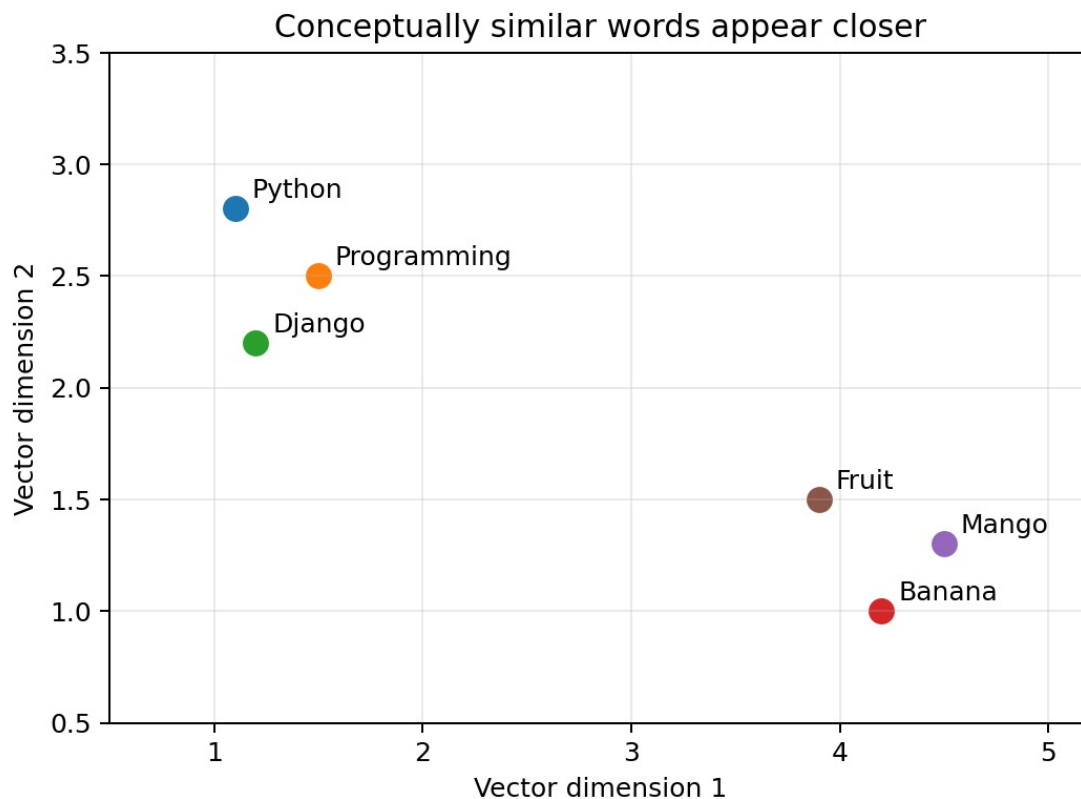
Storage and comparison: The vector can be stored in a vector database and compared with other vectors.

Important point

You normally do not interpret each individual number in an embedding. The meaning is distributed across the complete vector and becomes useful through distance or similarity calculations.

4. Semantic Vector Space

An embedding space is a multidimensional coordinate system. Items with similar meaning are positioned closer together, while unrelated items are generally farther apart.



The diagram uses only two dimensions for visualization. Real embedding models often produce hundreds or thousands of dimensions.

Dense vector

Most modern embeddings are dense vectors. This means that many vector positions contain meaningful non-zero values. Dense vectors are compact and effective for representing semantic information.

Vector dimension

Dimension means the number of values inside a vector. A model may generate vectors with 384, 768, 1,024, 1,536, or another fixed size. Every vector compared within the same index must normally have the same dimension.

5. Measuring Similarity

After creating embeddings, we compare vectors to estimate how similar their meanings are.

Metric	Meaning	Common interpretation
Cosine similarity	Compares the angle between two vectors	Closer to 1 generally means more similar
Euclidean distance	Measures straight-line distance	Smaller distance generally means more similar
Dot product	Multiplies and sums matching dimensions	Larger value can indicate stronger alignment

Cosine similarity intuition

Formula: $\text{cosine_similarity}(A, B) = (A \cdot B) / (||A|| \times ||B||)$

Hinglish intuition

Cosine similarity mainly vectors ki direction compare karti hai. Agar do vectors almost same direction mein point kar rahe hain, to unka semantic meaning bhi generally similar hota hai.

6. Embeddings in a RAG Pipeline

RAG stands for Retrieval-Augmented Generation. Embeddings allow a RAG system to retrieve relevant information before the language model generates an answer.

1. Load documents from sources such as text files, PDFs, webpages, or databases.
2. Split large documents into smaller chunks.
3. Generate an embedding for every chunk.
4. Store the embeddings and metadata in a vector database.
5. Convert the user question into an embedding using the same model.
6. Search for the nearest chunk vectors.
7. Send the retrieved chunks and the question to the LLM.
8. Generate a grounded answer using the retrieved context.

Critical rule

The document chunks and the user query should be embedded using the same embedding model. Different models may produce vectors with different dimensions and incompatible vector spaces.

Example metadata stored with a chunk

```
Document(
  page_content="Python is a high-level programming language...",
  metadata={
    "source": "python_intro.txt",
    "page": 1,
    "topic": "python"
  }
)
```

7. Python Example with LangChain

The following example shows the typical embedding workflow. Replace the model with one supported by your provider and environment.

```
from langchain_core.documents import Document
from langchain_huggingface import HuggingFaceEmbeddings

# 1. Create documents
documents = [
    Document(page_content="Python is a programming language."),
    Document(page_content="Django is a Python web framework."),
    Document(page_content="Mango is a tropical fruit."),
]

# 2. Load an embedding model
embeddings = HuggingFaceEmbeddings(
    model_name="sentence-transformers/all-MiniLM-L6-v2"
)

# 3. Generate vectors for document text
texts = [doc.page_content for doc in documents]
vectors = embeddings.embed_documents(texts)

print(len(vectors))      # Number of documents
print(len(vectors[0]))   # Embedding dimension
print(vectors[0][:5])    # First five values
```

Embedding a user query

```
query = "Which framework is used with Python?"
query_vector = embeddings.embed_query(query)

print(len(query_vector))
```

What happens next? The query vector is compared against the stored document vectors. The closest documents are returned as retrieval results.

8. Choosing an Embedding Model

Criterion	What to check
Language support	Use a model that performs well for the languages in

Criterion	What to check
	your documents and user queries.
Dimension	Higher dimensions may capture more detail but require more storage and computation.
Domain quality	General-purpose and domain-specific models may perform differently on legal, medical, code, or business data.
Speed and cost	Hosted APIs are convenient; local models may provide greater control and privacy.
Context length	Confirm how much text the model can embed in one request.
Retrieval evaluation	Select the model based on measured retrieval quality, not only popularity.

9. Common Mistakes

- Using different embedding models for indexing documents and embedding queries.
- Changing the embedding model without rebuilding the vector index.
- Creating chunks that are too large or too small for the information structure.
- Ignoring metadata such as source, page, topic, customer, or access permissions.
- Assuming a high similarity score always means the retrieved chunk is correct.
- Skipping retrieval evaluation with real user questions.
- Mixing vectors of different dimensions in the same vector index.

10. Interview-Ready Answers

Q: What is an embedding?

A: An embedding is a dense numerical vector that represents the semantic meaning and important features of data.

Q: Why are embeddings used in RAG?

A: They allow the system to compare a user query with document chunks by semantic similarity and retrieve relevant context.

Q: What is cosine similarity?

A: It measures the similarity between two vectors by comparing the angle between them.

Q: Why must the same model be used?

A: The model defines the vector space. Different models can produce incompatible dimensions and meanings.

Q: What is a vector database?

A: It is a database optimized to store embeddings and perform nearest-neighbor similarity search.

Key Takeaways

- Embeddings convert meaning into numerical vectors.
- Similar meanings generally produce nearby vectors.
- Similarity metrics help retrieve related information.
- RAG systems embed document chunks and user queries for semantic retrieval.
- Model consistency, chunk quality, metadata, and evaluation determine retrieval performance.

One-line summary

Embedding is the process of representing meaningful information as vectors so that machines can mathematically compare similarity.